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Designable Multiphase Nanocrystals Based on Phase Rearrangement

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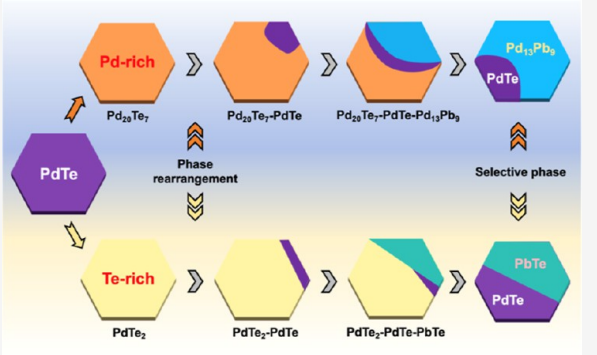
ABSTRACT: Phase engineering is a significant means of effectively enhancing catalytic performance. However, achieving continuous and precise phase control in intermetallic compounds remains a challenge, which has hindered the development of phase-dependent catalysis. In this work, we demonstrate continuous phase regulation and unveil the crucial role of phase rearrangement in the selective construction of nanocrystals. Through a mechanistic investigation, we uncovered a phase-dependent behavior governing the selective phase transformation during phase rearrangement. When a Pd-rich ($Pd_{20}Te_7$, the ratios of Pd/Te > 1) template was used, $Pd_{20}Te_7$ rearranged into PdTe. The Te vacancies generated in this process were occupied by Pb atoms, leading to the formation of a Pd−Pb alloy. Conversely, Te-rich template ($PdTe_2$, the ratios of Pd/Te < 1) rearranged to PdTe, with the resulting Pd vacancies being occupied by Pb atoms to form a Pb−Te alloy. The customized interface in PdTe−PbTe significantly enhanced its formic acid oxidation reaction (FAOR) activity. The catalyst achieved a mass activity of 4.17 A mg_{Pd}^{-1} and a specific activity of 9.12 mA cm^{-2}, surpassing commercial Pd/C by factors of 16.68 and 12.67, respectively. This work establishes a selective phase-regulation strategy to create customized nanostructures for diverse applications.



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■ INTRODUCTION

The catalytic activity of a material is governed by its electronic and atomic structure. In the field of controlled nanomaterial synthesis, the current approach typically requires the coordinated optimization of several key factors, including composition, morphology, size, dimensionality, and crystal facet.[1−6] For instance, the synthesis of size-controlled structures,[7−9] low-dimensional nanomaterials,[10−13] or the control of different crystal facets and exposing active sites[14−17] are currently the research hotspots for enhancing the activity of heterogeneous catalytic reactions. Crystal phase engineering, by altering the atomic arrangement and surface atomic structure of materials, can effectively regulate its structural properties, thereby endowing catalysts with enhanced performance.[18−23]

Phase rearrangement is a powerful strategy in crystal phase engineering, which can regulate the structure and properties of the materials. In recent years, phase rearrangement has been widely employed in a variety of materials, including metals,[24,25] alloys,[26−28] metal oxides,[29,30] and transition metal chalcogenides.[31−35] However, customized nanocrystalline structures based on phase rearrangement still lack selective and continuously refined control, which leads to insufficient diversity of crystal libraries and hinders true customization. Moreover, the mechanism of the induced phase rearrangement has not been elucidated. Therefore, achieving selective and continuous crystal phase control via phase rearrangement between different nanomaterials is of great significance for enriching the diversity of crystal libraries and further elucidating their synthetic mechanisms.

In this work, we constructed a series of heterostructures through crystal phase regulation and demonstrated the selective phase transformation via phase rearrangement. When a Pd-rich template was used, the structure evolved from $Pd_{20}Te_7$ to stable PdTe generating Te vacancies in the process of phase rearrangement, thereby allowing Pb atoms to occupy these sites and preferentially form Pd−Pb alloys. On the contrary, when the template was Te-rich, the structure evolved from $PdTe_2$ to stable PdTe, which generated Pd vacancies to form Pb−Te alloys. The formic acid oxidation reaction (FAOR) was used to probe the performance of phase-dependent heterogeneous electrocatalysts. The enhanced activity of PdTe−PbTe is attributed to the generated heterogeneous interface, which yielded remarkable mass and specific activity of 4.17 A mg_{Pd}^{-1}

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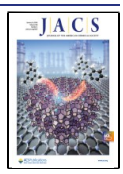

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and 9.12 mA cm^{-2}, respectively. These values are 16.68 and 12.67 times greater than those of state-of-the-art commercial Pd/C. The technique enables the selective design of distinct nanostructures, which can be used for various catalytic applications.

■ EXPERIMENTAL SECTION

Chemicals

Palladium(II) bis(acetylacetonate) ($Pd(acac)_2$, 99%), antimony(III) acetate ($Sb(CH_3COO)_3$, 99.9%), bismuth(III) acetate ($Bi(CH_3COO)_3$, 99%), and triethylene glycol (TEG, analytical reagent) were purchased from Aladdin, telluric (VI) acid (H_6TeO_6, 98%) was purchased from Sigma-Aldrich, indium(III) acetate ($In(CH_3COO)_3$, 99.9%) was purchased from Alfa Aesar, tin(II) acetate ($Sn(CH_3COO)_2$, 98%) was purchased from Zhengzhou Xipaike Technology Co. Ltd., and lead(II) acetate trihydrate ($Pb(CH_3COO)_2{\cdot}3H_2O$, 99.5%) was purchased from Xilong Scientific Co. Ltd. Polyvinylpyrrolidone (PVP, MW = 58,000) and citric acid (CA, 99.5%) were purchased from J&K. *N*,*N*-dimethylacetamide (DMAC, analytical reagent), benzyl alcohol (BP, analytical reagent), and ammonium chloride (NH_4Cl, 99.5%) were purchased from Sinopharm Chemical Reagent Co. Ltd. (Shanghai, China). All chemicals were used as received without further purification. The water (18 MΩ cm^{-1}) used in all experiments was prepared by passing it through an ultrapure purification system (Aqua Solutions).

Synthesis of PdTe/$Pd_{20}Te_7$/$PdTe_2$ NPs

The PdTe NPs were prepared via a wet chemical approach. $Pd(acac)_2$ (7.6 mg), H_6TeO_6 (5.8 mg), CA (30 mg), NH_4Cl (8 mg), PVP (50 mg), and DMAC (10 mL) were added into a capped vial (30 mL) and dispersed by ultrasonication for around 30 min. The mixture was heated to 180 °C and maintained for 2 h in an oil bath. The products were collected by centrifugation and washed with ethanol/acetone (1:9) one time and then dispersed in benzyl alcohol (2 mL) for further use. Other Pd−Te alloys were synthesized similarly by adjusting the amounts of the Te precursor and PVP. $Pd_{20}Te_7$ (2 mg H_6TeO_6 and 300 mg PVP) and $PdTe_2$ (12 mg H_6TeO_6 and 100 mg PVP) were obtained, respectively.

Synthesis of X-Introduced (X = In, Sn, Sb, Bi) PdTe

The benzyl alcohol solution (3 mL) of the X precursor ($In(CH_3COO)_3$ 2.2 mg, $Sn(CH_3COO)_2$ 2.4 mg, $Sb(CH_3COO)_3$ 2.4 mg, $Bi(CH_3COO)_3$ 3.1 mg), and as-synthesized Pd−Te alloys' homogeneous solution (2 mL) were mixed in a capped vial (30 mL). The mixture was ultrasonically treated for about 5 min, heated to 200 °C, and maintained for 3 h in an oil bath. The products were collected by centrifugation and washed with ethanol/acetone (1:9) three times.

Synthesis of Pb-Introduced Pd_xTe_y with Different Interfaces

In a typical ion exchange reaction, a desired amount (Table S1) of benzyl alcohol solution (3 mL) of the Pb precursor ($Pb(CH_3COO)_2{\cdot}3H_2O$) and an as-synthesized Pd−Te alloy homogeneous solution (2 mL) were mixed in a capped vial (30 mL). The mixture was ultrasonically treated for about 5 min and then heated to 200 °C and maintained for 3 h in an oil bath. The products were collected by centrifugation and washed three times with an ethanol/acetone (1:9 v/v) mixture.

Characterizations

The X-ray diffraction (XRD) patterns were collected on a SmartLab-SE powder diffractometer using a Cu radiation source (λ = 0.15406 nm). The concentrations of the samples were collected with an inductively coupled plasma optical emission spectrometer (ICP-OES, SPECTROBLUE FMX36, SPECTRO, Germany). Transmission electron microscopy (TEM) images were performed on a JEM-1400 TEM system at an accelerating voltage of 100 kV. The high-resolution transmission electron microscopy (HRTEM) images and the scanning transmission electron microscopy (STEM) mode were performed using a JEOL JEM-F200 at an accelerating voltage of 200 kV. The aberration-corrected HAADF−STEM imaging technique was acquired using a JEOL JEM-ARM300F2 at an accelerating voltage of 300 kV. X-ray photoelectron spectroscopy (XPS) spectra were obtained on an SSI S-Probe spectrometer. X-ray absorption fine structure (XAFS) data were measured at the National Synchrotron Radiation Research Center (NSRRC), Hsinchu, Taiwan, 300092. The in situ Fourier transform infrared spectroscopy (FTIR) measurement was conducted using a PerkinElmer Frontier FTIR Spectrometer equipped with a built-in MCT detector and unpolarized IR radiation at an incidence angle of ca. 70°. The electron paramagnetic resonance (EPR) spectroscopy was recorded on a Bruker EMX plus-9.5/12/P/L.

Electrochemical Measurements

Prior to electrochemical testing, the catalyst and carbon black (Vulcan XC-72) were separately dispersed in 5 mL of ethanol. The mixture was sonicated for 1 h, then collected by centrifugation, washed once with ethanol, and dried at room temperature. Finally, homogeneous catalyst inks were prepared by dispersing the catalysts in a mixture of 995 μL of 2-propanol and 5 μL of Nafion, followed by 30 min of sonication. The concentration of Pd for all catalysts was determined to be 0.4 mg mL^{-1} according to ICP-OES measurement. The loading amount of Pd onto the glassy carbon electrode (GCE) was 2 μg. All electrochemical measurements were carried out in a three-electrode configuration utilizing a CHI 660E workstation (Chenhua, Shanghai) and conducted at ambient temperature. A GCE with an area of 0.196 cm^2 served as the working electrode. A saturated calomel electrode (SCE) was used for the reference electrode, while the counter electrode was a graphite rod. The cyclic voltammetry (CV) tests were conducted in a 0.1 M $HClO_4$ solution at a sweep rate of 500 mV s^{-1}. Forward scans for the formic acid oxidation reaction (FAOR) were performed at a sweep rate of 50 mV s^{-1} in a solution of 0.1 M $HClO_4$ and 0.5 M HCOOH.

■ RESULTS AND DISCUSSION

Pb-introduced PdTe with distinct interfaces was prepared by a two-step method (see Experimental Section and Table S1 for details). Figure 1a reveals the schematic illustration of a continuous phase variation from 2.6% Pb−PdTe, PdTe−PbTe, to PdTe−PbTe−$Pd_{13}Pb_9$, driven by a continuous Te leaching process and a Pb introduction process. The structure of PdTe is maintained when Pb is first introduced. As the reaction continues, the structure evolves from PdTe to PbTe, leading to a coexisting two-phase structure. Finally, PdTe, PbTe, and $Pd_{13}Pb_9$ coexist. The powder X-ray diffraction (PXRD) patterns were collected to analyze the related crystal structures (Figures 1b, S1 and Table S2). 2.6% Pb−PdTe displays a hexagonal crystalline structure of PdTe, with lattice parameters of $a = b =$ 4.15 Å and c = 5.67 Å (PDF no. 97-064-8992). The pattern of PdTe−PbTe is composed of two groups of different diffraction peaks, indexed to hexagonal PdTe and cubic PbTe phases, and the crystal parameters of PbTe are $a = b = c$ = 6.46 Å (PDF no. 97-003-8295). Compared with PdTe−PbTe, PdTe−PbTe−$Pd_{13}Pb_9$ exhibits a decrease in PdTe peak intensity alongside an increase of PbTe peak intensity. Significantly, a group of new diffraction peaks emerged, which can be attributed to the monoclinic crystalline structure of $Pd_{13}Pb_9$, with lattice parameters of a = 15.592 Å, b = 9.053 Å, and c = 13.906 Å (PDF no. 97-064-8362).

Scanning electron microscopy-energy-dispersive X-ray spectroscopy (SEM-EDX) reveals (Figures 1c and S2) that the Pd/Te/Pb atomic ratios are 49.3/48.1/2.6, 47.1/43.2/9.7, and 38.9/34.9/26.2, which correspond to the 2.6% Pb−PdTe, PdTe−PbTe, and PdTe−PbTe−$Pd_{13}Pb_9$, respectively. The relative atomic content of Te was found to decline gradually, while the relative atomic content of Pb showed a constant accumulation, indicating the continuous Te leaching and Pb introduction during the process. Meanwhile, the results are further supported by inductively coupled plasma optical

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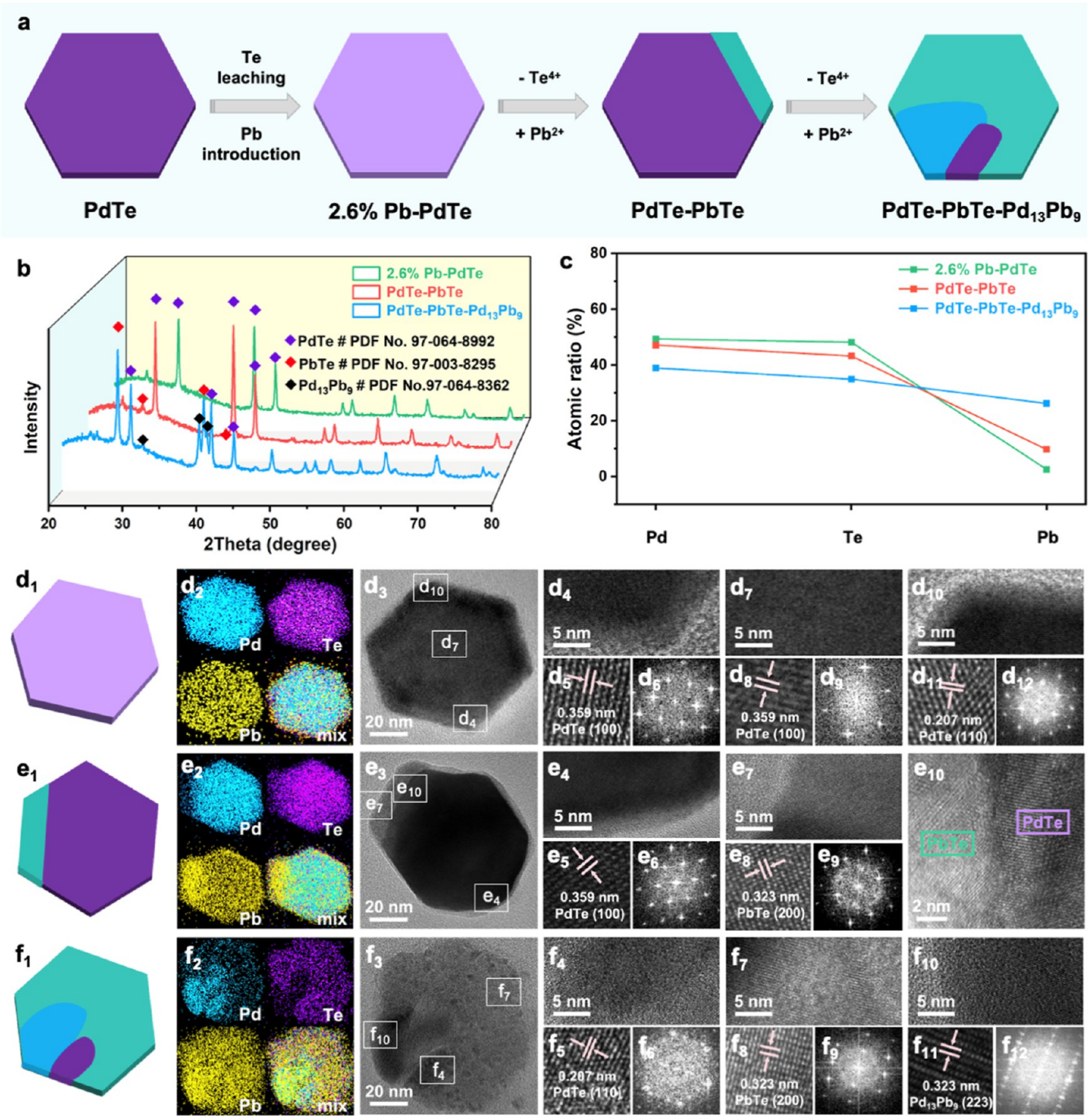

Figure 1. Morphology characterization and structure analysis of Pb-introduced PdTe with distinct interfaces. (a) Schematic illustration of a continuous phase variation between 2.6% Pb−PdTe, PdTe−PbTe, and PdTe−PbTe−$Pd_{13}Pb_9$, where PdTe, PbTe, and $Pd_{13}Pb_9$ are color-coded in purple, green, and blue, respectively. (b) Corresponding XRD patterns. (c) ICP-OES results of the atomic ratio. (d) Characterizations of 2.6% Pb−PdTe: (d_1) schematic model. (d_2) Elemental mapping image. (d_3) HRTEM image. (d_4,d_5) HRTEM image and enlarged HRTEM image from the selected area of figure d_3, and (d_6) corresponding FFT image. (d_7,d_8) HRTEM image and enlarged HRTEM image from the selected area of figure d_3 and (d_9) corresponding FFT image. (d_{10},d_{11}) HRTEM image and enlarged HRTEM image from the selected area of figure d_3 and (d_{12}) corresponding FFT image. (e) Characterizations of PdTe−PbTe. (f) Characterizations of PdTe−PbTe−$Pd_{13}Pb_9$.

emission spectrometry (ICP-OES) (Table S3). To precisely understand the elemental distribution in Pb-introduced PdTe with distinct interfaces (Figure 1d−f), corresponding schematic models were proposed to illustrate the crystal structure (Figure 1d_1−f_1). The element mapping images (Figures 1d_2−f_2 and S3) were further detected to reveal the distributions of Pd (blue), Te (purple), and Pb (yellow), showing that Pb is predominantly located at the outer edges of the nanoplate. As the Pb content increases, PdTe−PbTe exhibits a nanoplate with one side predominantly enriched in Pb. Finally, Pd, Te, and Pb reveal regional gathering when the atomic ratio of Pb is 26.2.

Furthermore, the microscopic morphology was characterized by TEM and HAADF (Figures S4 and 1d_3−f_3), confirming that all three samples consist of hexagonal nanoplates. The HRTEM images from selected areas (Figure 1d_4−f_4,d_7−f_7,d_{10},f_{10}) and their corresponding magnified views reveal the lattice spacings

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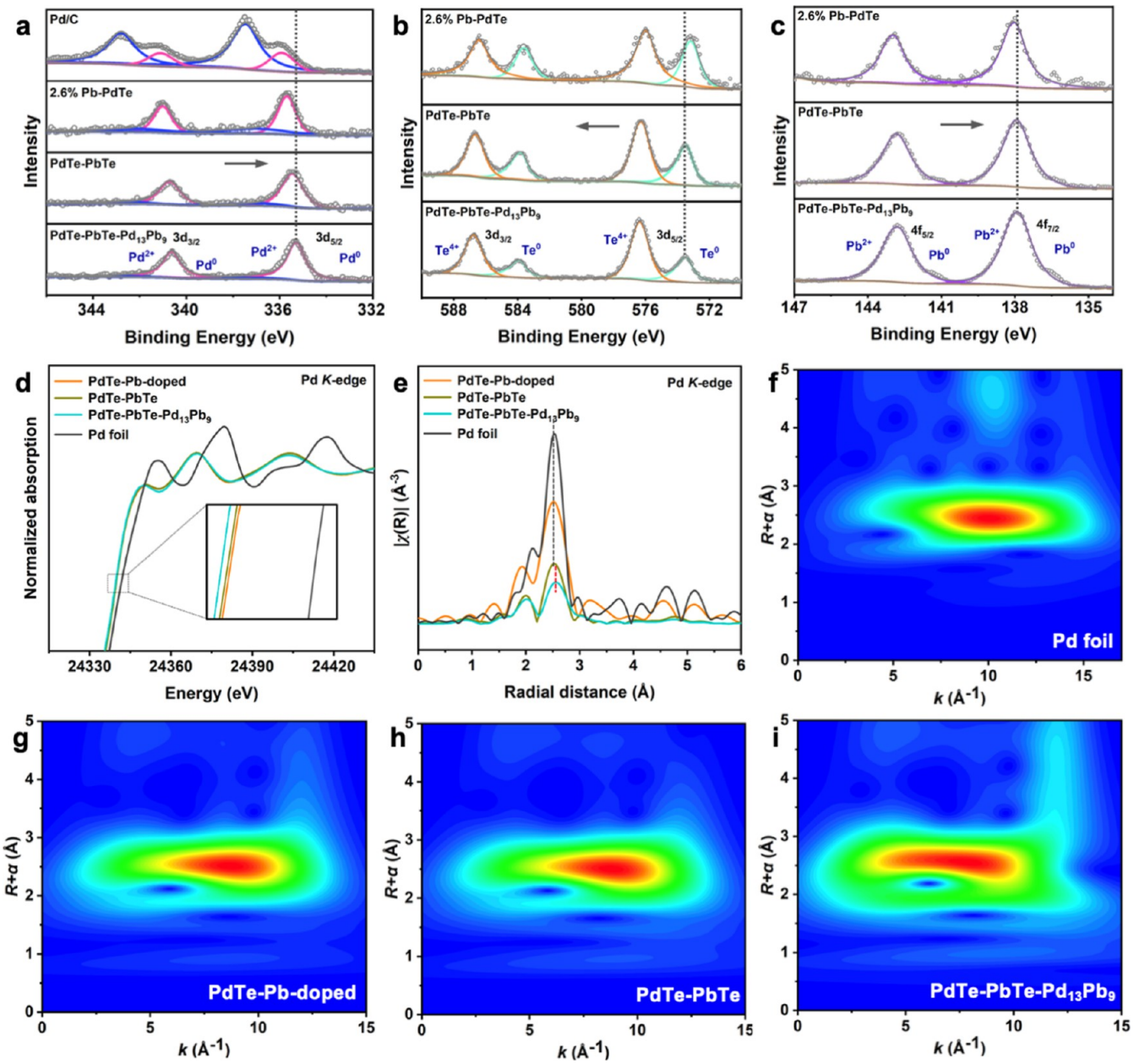

Figure 2. Structure analysis of a series of Pb-introduced PdTe with distinct interfaces. XPS spectra of (a) Pd 3d, (b) Te 3d, and (c) Pb 4f. (d) Pd K-edge XANES spectra of 2.6% Pb−PdTe, PdTe−PbTe, PdTe−PbTe−$Pd_{13}Pb_9$, and commercial Pd/C. (e) FT-EXAFS spectra at Pd K-edge. (f−i) Wavelet transform of Pd foil, 2.6% Pb−PdTe, PdTe−PbTe, and PdTe−PbTe−$Pd_{13}Pb_9$.

of the different phases. For 2.6% Pb−PdTe, the interplanar spacings of 0.359 and 0.207 nm are indexed to the (100) and (110) facets (Figure 1d_5,d_8,d_{11}), respectively. The fast Fourier transform (FFT) patterns further confirm the hexagonal structure of PdTe (Figure 1d_6,d_9,d_{12}). Meanwhile, the interplanar spacings of PdTe−PbTe are 0.222 and 0.323 nm, which are attributed to the (102) facets of PdTe and (200) facets of PbTe, respectively (Figure 1e_5,e_8). The FFT patterns likewise verify the corresponding crystalline structures of both phases (Figure 1e_6,e_9). The interface between PdTe and PbTe (Figure 1e_{10}) further proves the coexistence of two-phase. The interplanar spacings of PdTe−PbTe−$Pd_{13}Pb_9$ are 0.207, 0.323, and 0.323 nm, which can be ascribed to the (110) facets of PdTe, (200) facets of PbTe, and (223) facets of $Pd_{13}Pb_9$, respectively (Figure 1f_5,f_8,f_{11}), thereby confirming the three-phase coexistence. FFT patterns (Figure 1f_6,f_9,f_{12}) also verify the corresponding crystalline structures.

To further study the electronic structure of Pb-introduced PdTe with distinct interfaces, X-ray photoelectron spectroscopy (XPS) was employed. Remarkably, Pd predominantly exists in the metallic state across all three specimens. From 2.6% Pb−PdTe, PdTe−PbTe, to PdTe−PbTe−$Pd_{13}Pb_9$, the peak of Pd 3d (Figure 2a) reveals a tendency to shift toward lower binding energy, indicating Pd obtains electrons in this process. The Te 3d peaks (Figure 2b) exhibit a shift to higher binding energy, suggesting an increase in the oxidation state. Conversely, the Pb 4f peaks (Figure 2c) display a negative binding energy shift, attributed to electron transfer from Te to Pb during Pb−Te bond formation. Moreover, the electronic structure of the Pd species was further probed by X-ray absorption near-edge spectroscopy (XANES), which reveals a shift to a lower energy

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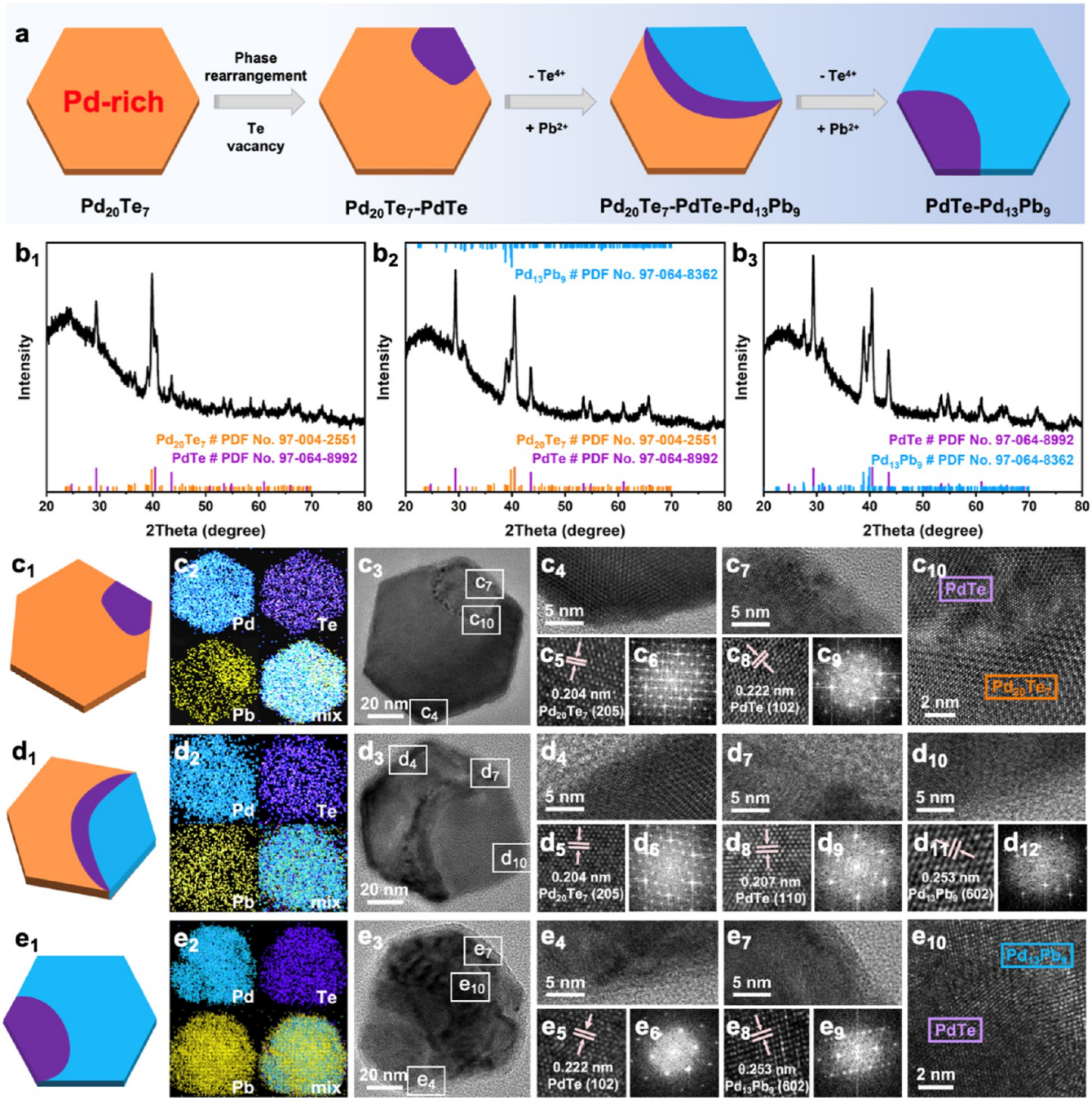

Figure 3. Morphology characterization and structure analysis of Pb-introduced $Pd_{20}Te_7$ with distinct interfaces. (a) Schematic illustration of a continuous phase variation among $Pd_{20}Te_7$−PdTe, $Pd_{20}Te_7$−PdTe−$Pd_{13}Pb_9$, and PdTe−$Pd_{13}Pb_9$, where $Pd_{20}Te_7$, PdTe, and $Pd_{13}Pb_9$ are color-coded in orange, purple, and blue, respectively. (b_1−b_3) Corresponding XRD patterns. (c) Characterizations of $Pd_{20}Te_7$−PdTe: (c_1) schematic model. (c_2) Elemental mapping image. (c_3) HRTEM image. (c_4,c_5) HRTEM image and enlarged HRTEM image from the selected area of figure c_3 and (c_6) corresponding FFT image. (c_7,c_8) HRTEM image and enlarged HRTEM image from the selected area of figure c_3 and (c_9) corresponding FFT image. (c_{10}) Enlarged HRTEM image of the interface from the selected area of figure c_3. (d) Characterizations of $Pd_{20}Te_7$−PdTe−$Pd_{13}Pb_9$. (e) Characterizations of PdTe−$Pd_{13}Pb_9$.

position compared with Pd foil (Figures 2d and S5), in agreement with the XPS findings.

The Fourier transform k^3-weighted extended X-ray absorption fine structure spectroscopy (EXAFS) spectra of the Pd K-edge were conducted to elucidate the coordination structure. As shown in Figure 2e, the Pd foil exhibits an intense Pd−Pd peak at 2.51 Å. From 2.6% Pb−PdTe and PdTe−PbTe to PdTe−PbTe−$Pd_{13}Pb_9$, the spectra show an attenuation of peak intensity, indicating a decreasing Pd−Pd coordination number. This is attributed to the breaking of Pd−Te bonds during the concurrent formation of Pb−Te bonds. In contrast, the position of the Pd−Te peak for 2.6% Pb−PdTe was basically unchanged (2.51 Å). However, PdTe−PbTe and PdTe−PbTe−$Pd_{13}Pb_9$ displayed a positive shift corresponding to the Pd−Te peak at 2.53 Å and the Pd−Pd/Te/Pb peak at 2.55 Å, respectively. The observed shift is indicative of pronounced tensile strain in the Pd−Te bonds. The wavelet transform analysis (Figure 2f−i) exhibits the peak intensity of 2.6% Pb−PdTe, PdTe−PbTe, and

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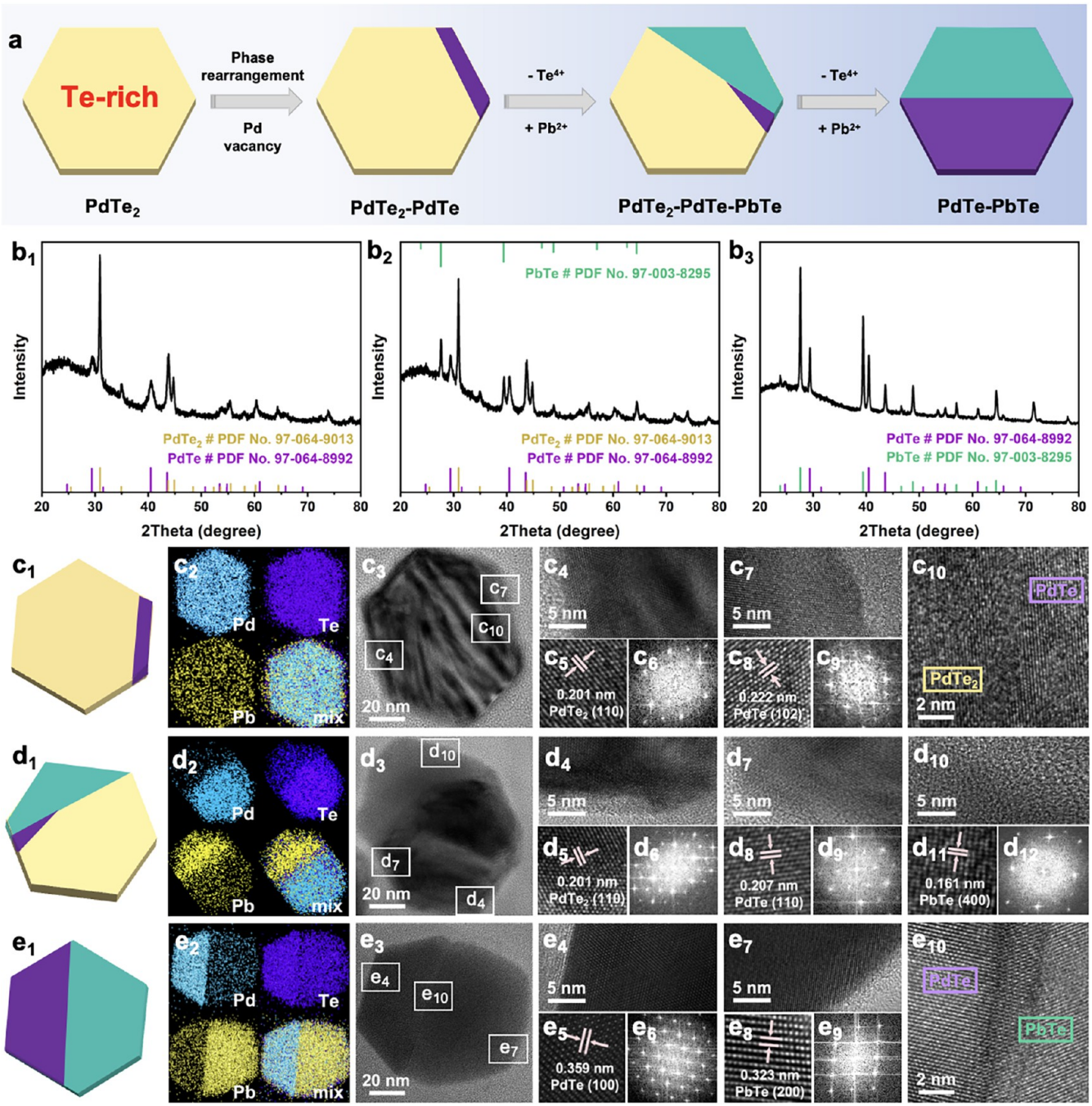

Figure 4. Morphology characterization and structure analysis of Pb-introduced $PdTe_2$ with distinct interfaces. (a) Schematic illustration of a continuous phase variation among $PdTe_2$−PdTe, $PdTe_2$−PdTe−PbTe, and PdTe−PbTe, where $PdTe_2$, PdTe, and PbTe are color-coded in yellow, purple, and green, respectively. (b_1−b_3) Corresponding XRD pattern. (c) Characterizations of $PdTe_2$−PdTe: (c_1) schematic model. (c_2) Elemental mapping image. (c_3) HRTEM image. (c_4,c_5) HRTEM image and enlarged HRTEM image from the selected area of figure c_3 and (c_6) corresponding FFT image. (c_7,c_8) HRTEM image and enlarged HRTEM image from the selected area of figure c_3 and (c_9) corresponding FFT image. (c_{10}) Enlarged HRTEM image of the interface from the selected area of figure c_3. (d) Characterizations of $PdTe_2$−PdTe−PbTe. (e) Characterizations of PdTe−PbTe.

PdTe−PbTe−$Pd_{13}Pb_9$ at 8.87 $Å^{-1}$, 8.76 $Å^{-1}$, and 7.24 $Å^{-1}$, respectively. These values are progressively lower than those of Pd foil (10.09 $Å^{-1}$), thus confirming the different degrees of tensile strain in the three samples.

Notably, we found that using Pd-rich or Te-rich templates enables the selective synthesis of distinct phases through a phase rearrangement process (see Experimental Section and Table S1 for details). Figure 3a reveals the continuous phase variation from $Pd_{20}Te_7$, $Pd_{20}Te_7$−PdTe, $Pd_{20}Te_7$−PdTe−$Pd_{13}Pb_9$, to PdTe−$Pd_{13}Pb_9$. When $Pd_{20}Te_7$ was used as the template, the structure first evolved from $Pd_{20}Te_7$ to PdTe via phase rearrangement, a process accompanied by the generation of Te vacancies. Compared to the original $Pd_{20}Te_7$, the electron paramagnetic resonance (EPR) spectrum of $Pd_{20}Te_7$−PdTe exhibits a significantly enhanced signal peak at g = 2.02, suggesting the generation of Te vacancies (Figure S6). This finding was further validated by an aberration-corrected HAADF−STEM (AC−HAADF−STEM) image (Figure S7).

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Pb then occupies these vacancies to form $Pd_{13}Pb_9$. As the reaction continues, $Pd_{20}Te_7$ transforms completely into PdTe, resulting in a final structure where PdTe and $Pd_{13}Pb_9$ coexist. The PXRD pattern of $Pd_{20}Te_7$−PdTe displays two groups of different diffraction peaks, corresponding to the hexagonal crystalline structure of $Pd_{20}Te_7$ and PdTe. The lattice parameters of $Pd_{20}Te_7$ are $a = b = 11.80$ and $c = 11.17$ Å (PDF no. 97−004−2551). Compared with $Pd_{20}Te_7$−PdTe, $Pd_{20}Te_7$−PdTe−$Pd_{13}Pb_9$ shows a decrease in $Pd_{20}Te_7$ peak intensity and an increase of PdTe peak intensity. Meanwhile, a group of new diffraction peaks was observed, corresponding to $Pd_{13}Pb_9$. The pattern of PdTe−$Pd_{13}Pb_9$ displays two groups of diffraction peaks, ascribed to PdTe and $Pd_{13}Pb_9$. No $Pd_{20}Te_7$ peak was found in PdTe−$Pd_{13}Pb_9$, proving that $Pd_{20}Te_7$ was completely consumed and rearranged into PdTe. This observation is supported by the corresponding PXRD patterns (Figures 3b_1−b_3, S8 and Table S2). As SEM-EDX and ICP-OES (Figure S9 and Table S3) reveal a decrease in the relative atomic content of Te, the relative atomic content of Pb displays a constant accumulation. To better understand the crystal structure, the corresponding schematic models are presented in Figure 3c_1−e_1. Moreover, element mapping images (Figure 3c_2−e_2) are shown to reveal the distributions of Pd (blue), Te (purple), and Pb (yellow). Notably, Pb initially distributes predominantly around the position of PdTe, a consequence of the Te vacancies generated during phase rearrangement.

The microscopic morphology was characterized by TEM and HAADF (Figures S10 and 3c_3−e_3), confirming that all samples adopt a consistent hexagonal nanoplate morphology. The HRTEM images from selected areas (Figure 3c_4−e_4,c_7−e_7,d_{10}) and corresponding enlarged images reveal the interplanar spacing of different phases. The interplanar spacings of $Pd_{20}Te_7$−PdTe are 0.204 and 0.222 nm, which can be attributed to the (205) facets of $Pd_{20}Te_7$ and (102) facets of PdTe (Figure 3c_5,c_8), respectively. The FFT patterns (Figure 3c_6,c_9) further verify these crystalline structures. The interface between $Pd_{20}Te_7$ and PdTe (Figure 3c_{10}) further proves the two-phase coexistence. Figure 3d_5,d_8,d_{11} reveals the $Pd_{20}Te_7$−PdTe−$Pd_{13}Pb_9$ with the lattice spacings of 0.204 nm, 0.207 and 0.253 nm, which are assigned to the (205) facets of $Pd_{20}Te_7$, (110) facets of PdTe, and (602) facets of $Pd_{13}Pb_9$, respectively. The FFT patterns (Figure 3d_6,d_9,d_{12}) further confirm the corresponding crystalline structure. Meanwhile, the interplanar spacings of PdTe−$Pd_{13}Pb_9$ are 0.222 and 0.253 nm, corresponding to the (102) facets of PdTe and (602) facets of $Pd_{13}Pb_9$ (Figure 3e_5,e_8), respectively. The FFT patterns (Figure 3e_6,e_9) likewise verify the corresponding crystalline structure. The interface between PdTe and $Pd_{13}Pb_9$ (Figure 3e_{10}) further proves the two-phase coexistence. Notably, density functional theory (DFT) calculations show that the final phases (PdTe and $Pd_{13}Pb_9$) are thermodynamically more stable than $Pd_{20}Te_7$, with a total-energy decrease of −2.52 eV/f.u. This indicates a thermodynamic driving force for the experimentally observed phase rearrangement, while kinetic barriers and the detailed transformation pathway are not addressed in this work (Figure S11).

In sharp contrast, when $PdTe_2$ was used as the template, a continuous phase variation from $PdTe_2$, $PdTe_2$−PdTe, and $PdTe_2$−PdTe−PbTe, to PdTe−PbTe was observed (Figure 4a). The structure also undergoes an evolution from $PdTe_2$ to PdTe via phase rearrangement with the concomitant generation of Pd vacancies during the reaction. The EPR spectrum of $PdTe_2$−PdTe shows a more intense signal at $g = 2.02$ compared to that of pristine $PdTe_2$, consistent with the creation of Pd vacancies in this process (Figure S12). The result is directly confirmed by the image of AC−HAADF−STEM (Figure S13). Pb atoms subsequently occupy these vacancy sites, leading to the formation of PbTe. As the reaction proceeds, $PdTe_2$ transforms to PdTe completely and PdTe and PbTe coexist finally. The PXRD pattern of $PdTe_2$−PdTe displays two groups of different diffraction peaks, corresponding to the hexagonal crystalline structure of $PdTe_2$ and PdTe. The lattice parameters of $PdTe_2$ were determined to be $a = b = 4.04$ Å and $c = 5.13$ Å (JCPDS no. 97-064-9013). Compared with $PdTe_2$−PdTe, $PdTe_2$−PdTe−PbTe exhibits an increase in the PdTe peak intensity along with the appearance of a new group of PbTe peaks. The pattern of PdTe−PbTe shows two groups of diffraction peaks, ascribed to PdTe and PbTe. No $PdTe_2$ peaks were found in PdTe−PbTe, confirming that $PdTe_2$ was rearranged to PdTe completely. These phase transformations are corroborated by the PXRD patterns (Figures 4b_1−b_3, S14 and Table S2). Both SEM-EDX and ICP-OES analyses (Figure S15 and Table S3) display a decrease in the relative atomic content of Te and an increase of the relative atomic content of Pb. Schematic models are presented to help understand the crystal structure (Figure 4c_1−e_1). The element mapping images (Figure 4c_2−e_2) further reveal the distribution of Pd (blue), Te (purple), and Pb (yellow).

Furthermore, TEM and HAADF (Figures S16 and 4c_3−e_3) were employed to characterize the microscopic morphology. The HRTEM images from selected areas (Figure 4c_4−e_4,c_7−e_7,d_{10}) and corresponding enlarged images reveal the lattice spacing of different phases. The interplanar spacings of $PdTe_2$−PdTe are 0.201 and 0.222 nm, which can be attributed to the (110) facets of $PdTe_2$ and (102) facets of PdTe, respectively (Figure 4c_5,c_8). The corresponding FFT patterns further confirm the identified crystal structure (Figure 4c_6,c_9). Moreover, the interface between $PdTe_2$ and PdTe (Figure 4c_{10}) further proves the two-phase coexistence. $PdTe_2$−PdTe−PbTe shows lattice spacings of 0.201 nm, 0.207 nm, and 0.161 nm, which can be referred to the (110) facets of $PdTe_2$, (110) facets of PdTe, and (400) facets of PbTe, respectively (Figure 4d_5, 4d_8, 4d_{11}). The result proves the coexistence of three crystal phases. The FFT patterns (Figure 4d_6,d_9,d_{12}) further confirm the corresponding crystalline structure. The interplanar spacings of PdTe−PbTe are 0.359 and 0.323 nm, corresponding to the (100) facets of PdTe and (200) facets of PbTe, respectively (Figure 4e_5,e_8). The FFT patterns confirm it as well (Figure 4e_6,e_9). The interface between PdTe and PbTe (Figure 4e_{10}) further verifies the two-phase coexistence. Simultaneously, we compared the DFT total energies of the initial ($PdTe_2$) and final (PdTe + PbTe) states to assess the thermodynamic feasibility. The products are lower by 0.53 eV per fu, indicating a thermodynamic driving force for the transformation (Figure S17).

Furthermore, the roles of the introduced elements and solvent in determining the final crystal structures were studied. Pb is crucial for the formation of the phase structure. When In, Sn, Sb, or Bi was used instead of Pb with a PdTe template, the original PdTe structure remained unchanged (Figure S18). Conversely, when BP was replaced with TEG or PEA, a continuous phase variation from PdTe, PdTe−PbTe, to PdTe−PbTe−$Pd_{13}Pb_9$ was still observed (Figures S19 and S20). Notably, a similar progression of phase variations was achieved when Pd_8Te_3 (Pd/Te > 1) served as the template, evolving from Pd_8Te_3, Pd_8Te_3−PdTe, Pd_8Te_3−PdTe−$Pd_{13}Pb_9$, to PdTe−$Pd_{13}Pb_9$ (Figure

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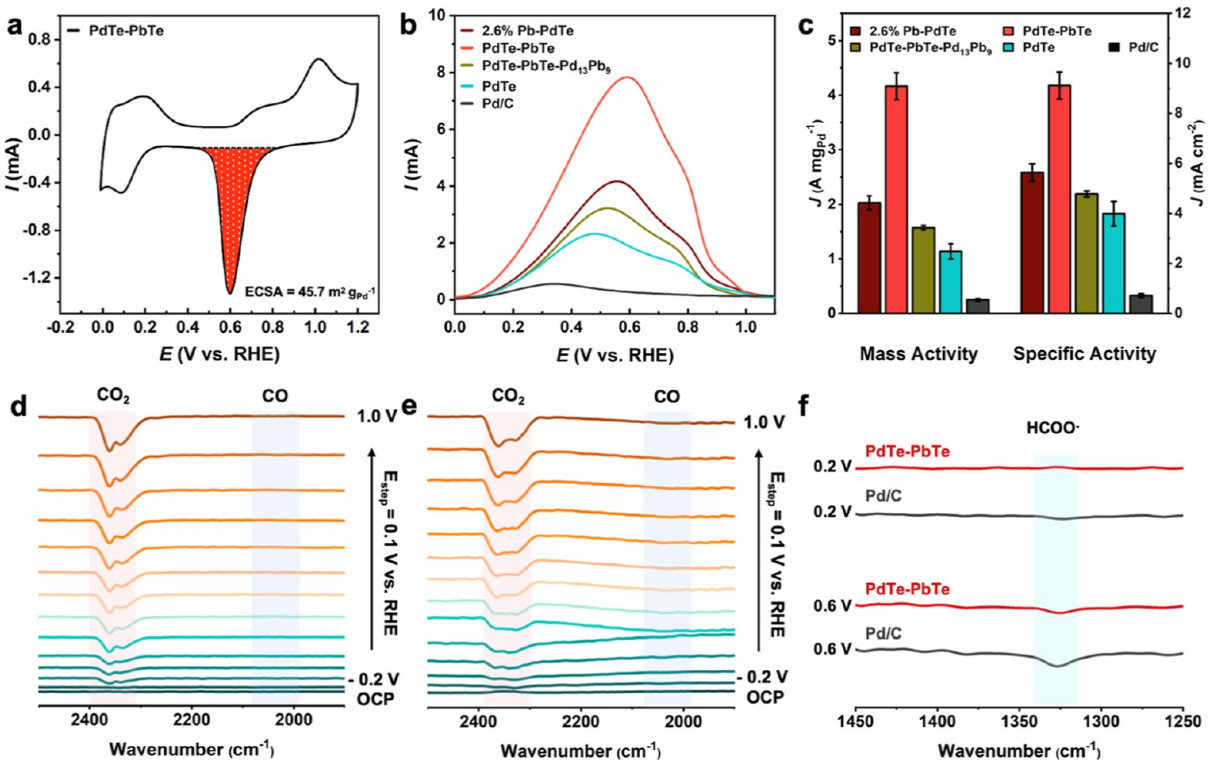

Figure 5. FAOR evaluation of Pb-introduced PdTe with distinct interfaces. (a) CV curves of PdTe−PbTe in 0.1 M $HClO_4$. (b) Forward scan curves of different catalysts toward FAOR with a sweep rate of 50 mV s^{-1} in 0.1 M $HClO_4$ + 0.5 M HCOOH electrolyte. (c) Graphical charts of the normalized mass activities and specific activities of 2.6% Pb−PdTe, PdTe−PbTe, PdTe−PbTe−$Pd_{13}Pb_9$, PdTe, and commercial Pd/C. In situ FTIR spectra of (d) PdTe−PbTe and (e) commercial Pd/C. (f) In situ ATR-FTIR spectra of PdTe−PbTe and commercial Pd/C measured at 0.2 and 0.6 V vs RHE.

S21). The result further confirms the universality of our proposed phase-regulation mechanism.

FAOR was chosen as a proof of concept to illustrate the performance of phase-dependent heterogeneous electrocatalysts. Before the electrochemical measurements, the catalysts were loaded onto Vulcan XC-72R carbon with 20 wt % Pd content by sonicating (see Experimental Section and Figure S22 for details). The CV curves of the catalysts were carried out in 0.1 M $HClO_4$ solution at the scan rate of 50 mV s^{-1} to measure the electrochemically active surface area (ECSA). Taking PdTe−PbTe for instance, ECSA of PdTe−PbTe is 45.7 m^2 g_{Pd}^{-1} (Figure 5a), which was calculated by the surface redox for the surface area of Pd (Pattern-filled section), where the ECSAs of 2.6% Pb−PdTe, PdTe−PbTe−$Pd_{13}Pb_9$, PdTe, and commercial Pd/C are 36.2, 32.9, 28.5, and 34.8 m^2 g_{Pd}^{-1}, respectively (Figure S23). The FAOR forward scan curves measured in 0.1 M $HClO_4$ + 0.5 M HCOOH solution reveal a distinct direct oxidation peak of PdTe−PbTe, indicating that the direct pathway is dominant (Figure 5b). Furthermore, the mass activity (MA) and specific activity (SA) of different electrocatalysts were calculated by normalizing the forward scan current to the loading amount of Pd and the ECSA. Notably, the MA exhibits a volcano-type trend. PdTe−PbTe displays the highest MA of 4.17 A mg_{Pd}^{-1}, which is 16.68 times higher than that of commercial Pd/C (0.25 A mg_{Pd}^{-1}). The MAs of 2.6% Pb−PdTe, PdTe−PbTe−$Pd_{13}Pb_9$, and PdTe were determined to be 2.03, 1.57, and 1.14 A mg_{Pd}^{-1}, respectively. Mirroring the MA trend, PdTe−PbTe also shows the highest SA of 9.12 mA cm^{-2}, which is 12.67 times higher than that of commercial Pd/C (0.72 mA cm^{-2}). The SAs of 2.6% Pb−PdTe, PdTe−PbTe−$Pd_{13}Pb_9$, and PdTe were determined to be 5.64, 4.78, and 3.99 mA cm^{-2}, respectively (Figures 5c, S24, Tables S4 and S5). These results demonstrate that the catalytic performance is governed by the combination of the crystal phase and interfacial effects. This conclusion is further confirmed by a control experiment using pure PbTe, which exhibits a negligible MA of only 0.07 A mg_{Pb}^{-1} (Figure S25). In contrast, the FAOR performance of our synthesized PdTe−PbTe (4.17 A mg_{Pd}^{-1}) is far superior to that of either PdTe (1.14 A mg_{Pd}^{-1}) or PbTe (0.07 A mg_{Pb}^{-1}) alone, which indicates that the enhanced performance of PdTe−PbTe is attributed to the interface formed between the two phases. Furthermore, the stability of PdTe−PbTe was detected by TEM and SEM-EDX (Figure S26) after 5000 CV cycles. The nanostructure remains intact, while the Pb content reveals no measurable decrease compared to the original catalyst.

To further understand the reaction mechanism, the in situ Fourier transform infrared spectroscopy (FTIR) technique was carried out to detect reaction intermediates. The characteristic peaks of CO_2 at 2341 cm^{-1} can be obviously observed in PdTe−PbTe, whereas no peaks for CO are detected at 2030 cm^{-1}, and confirms that the reaction follows the direct formic acid oxidation pathway (Figure 5d). However, the characteristic peaks of CO_2 and the slight characteristic peaks of CO can also be observed in commercial Pd/C, suggesting the existence of an indirect path (Figure 5e). Additionally, the in situ attenuated total reflection (ATR)-FTIR technique was employed to further investigate the reactive adsorption intermediates. The bridge adsorption of $HCOO^-$ intermediates effectively reduces the adsorption of CO intermediates, thereby suppressing the indirect pathway. The adsorption capacity of the $HCOO^-$ intermediates reaches the maximum value at 0.6 V vs RHE (Figure 5f). The construction of alloy interfaces is pivotal for enhanced electrocatalytic performance. The interface of PdTe−

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PbTe provides more active sites, which can effectively enhance the adsorption of reaction intermediates and reduce the activation barrier of the reaction, inducing excellent performance in FAOR.

■ CONCLUSIONS

To summarize, we have successfully achieved continuous regulation of Pb-introduced crystal phases; eight heterostructures with crystal phase dependence were synthesized by changing the type of template and introducing contents. In addition, we propose a novel mechanism for customizing nanostructures based on phase-dependent vacancy engineering. The phase regulation can be selectively controlled when altering the phase of the templates. This selectivity resulted from the different types of vacancies generated in phase rearrangement between $Pd_{20}Te_7$ to PdTe and $PdTe_2$ to PdTe. Therefore, the Pd−Pb alloy and Pb−Te alloy can be selectively obtained ultimately. Our work emphasized the key role of phase rearrangement strategy in guiding the selective synthesis of heterostructures and enriched the multiphase nanocrystals library of materials with remarkable characteristics.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.5c14589.

Experimental data, including PXRD patterns of PdTe NPs, $Pd_{20}Te_7$ NPs, $PdTe_2$ NPs, Pd_8Te_3 NPs, Pd_8Te_3−PdTe, Pd_8Te_3−PdTe−$Pd_{13}Pb_9$, PdTe−$Pd_{13}Pb_9$, and PbTe; PXRD patterns; SEM-EDX spectra and TEM images of 2.6% Pb−PdTe, PdTe−PbTe, PdTe−PbTe−$Pd_{13}Pb_9$, $Pd_{20}Te_7$−PdTe, $Pd_{20}Te_7$−PdTe−$Pd_{13}Pb_9$, PdTe−$Pd_{13}Pb_9$, $PdTe_2$−PdTe, $PdTe_2$−PdTe−PbTe, and PdTe−PbTe; elemental mapping images of PdTe NPs; full normalized Pd K-edge XANES spectra; CVs of 2.6% Pb−PdTe, PdTe−PbTe, PdTe−PbTe−$Pd_{13}Pb_9$, PdTe, and commercial Pd/C; parallel forward scan curves of 2.6% Pb−PdTe, PdTe−PbTe, PdTe−PbTe−$Pd_{13}Pb_9$, PdTe, and commercial Pd/C toward FAOR for error range analysis; EPR spectra of $Pd_{20}Te_7$−PdTe, $Pd_{20}Te_7$, $PdTe_2$−PdTe, and $PdTe_2$; AC-HAADF-STEM images of $Pd_{20}Te_7$−PdTe and $PdTe_2$−PdTe; energy variation between the phase variation from $Pd_{20}Te_7$ to PdTe−$Pd_{13}Pb_9$ and $PdTe_2$ to PdTe−PbTe; TEM images of commercial Pd/C; synthetic parameters used with $Pb(CH_3COO)_2{\cdot}3H_2O$ and the composition of the corresponding products; unit-cell data and structure result; SEM-EDX and ICP-OES of Pb-introduced Pd_xTe_y with different interfaces; FAOR peak currents; mass activities; specific activities; and ECSAs of different catalysts (PDF)

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Notes

The authors declare no competing financial interest.

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